TEXORA · SESSIONAL NOTES · LAB EXPERIMENT SHEET

Data Mining Tools

Finding the patterns a mill's own records already contain: association rules mined from stores issue data in this lab — support and confidence calculated exactly — and the clustering and classification tools that complete the data miner's kit.

TE-3208 · SOFTWARE AND AI IN TEXTILE MANUFACTURING SESSIONAL · EXPERIMENT 8 · LEVEL 3 TERM 2

1. Aim

To mine a small transaction dataset with the association-rule method — computing support and confidence by hand and with a data-mining tool — and to state where association, clustering and classification each apply in textile manufacturing data.

2. Concepts you need

- **Data mining** is the search for patterns that are *valid* (they hold beyond the sample), *novel* (not already known) and *useful* (someone can act on them) in large records. It differs from reporting: a report answers the question you asked; mining finds the question you should have asked.
- **Association rules** have the form X → Y ("transactions containing X tend to contain Y"). Two measures grade every rule: **support** = the fraction of all transactions containing X and Y together — how much evidence exists; **confidence** = of the transactions containing X, the fraction also containing Y — how reliable the rule is. A rule needs both: high confidence on tiny support is an anecdote, not a pattern.
- **Thresholds:** the analyst sets a minimum support and minimum confidence; the tool (classically the Apriori algorithm and its descendants) enumerates the itemsets that pass. Raising min-support prunes rare coincidences; it can also prune rare-but-critical patterns, so thresholds are engineering judgement, not ritual.
- **Clustering** groups unlabelled records by similarity — e.g. sorting a season's fabric faults into families, or grouping looms by their stoppage fingerprints (Experiment 5's data) so maintenance attacks one family at a time. Unlike classification (Experiment 7), nobody supplies the answers in advance.
- **The tools:** mining is done at three levels — spreadsheet pivot/filter work for small sets; dedicated GUI mining environments (open-source workbenches such as WEKA or KNIME are the usual

teaching tools) for prepared datasets; and programming libraries for production systems. The tool changes; support, confidence and the train/test discipline do not.

- **Data preparation dominates:** most of any real mining project is cleaning — merging machine logs with different count conventions, removing duplicate lots, aligning dates. A pattern mined from dirty data is a confident error.
- **Correlation is not cause:** a rule says two things travel together in the records. Acting on it (stocking items together, changing a setting) is safe when the action is cheap and reversible; claiming one item *causes* the other needs evidence mining alone cannot give.

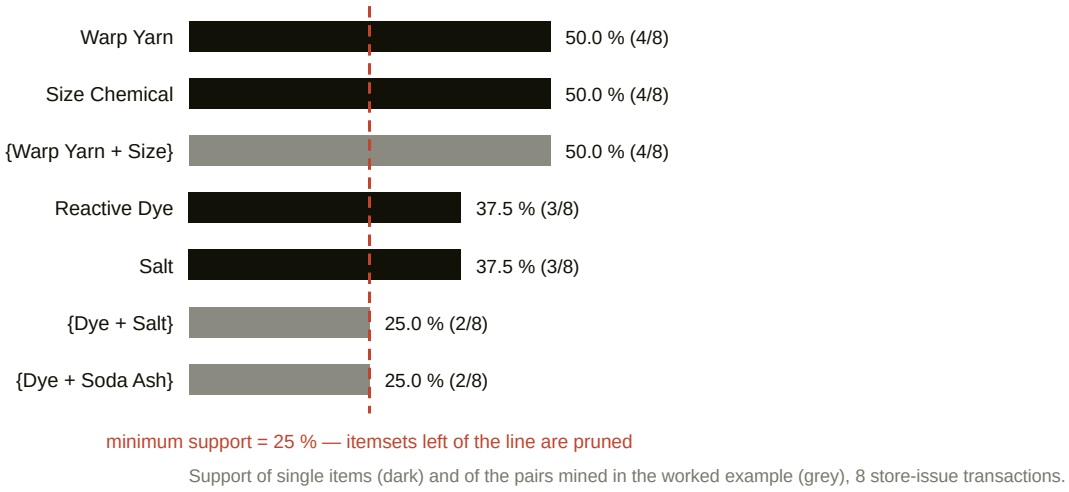


Figure 1 — Support chart for the worked example. Every bar at or right of the threshold line survives the min-support test; the pairs are then graded by confidence before any rule is believed.

3. Lab procedure

1. Enter the eight store-issue transactions of the worked example into the mining tool (or build the count table in a spreadsheet).
2. Compute every single-item support by hand; verify the tool's figures against Figure 1.
3. Set minimum support 25 % and minimum confidence 60 %; run the association mining and list the surviving rules.
4. Compute the confidence of each surviving rule by hand from the counts and reconcile with the tool.
5. Interpret the two strongest rules as stores policy: what should be stocked, kitted or reordered together, and what evidence would you check before acting?
6. Repeat with minimum support raised to 40 %: record which rules vanish, and state in one sentence what that teaches about threshold choice.

4. Worked example — association rules from the mill store

Eight issue transactions (one day's requisitions from the mill store). Items: warp yarn, weft yarn, size chemical, reactive dye, salt, soda ash.

TRANS.	ITEMS ISSUED TOGETHER
T1	Warp Yarn, Weft Yarn, Size
T2	Warp Yarn, Size
T3	Reactive Dye, Salt, Soda Ash
T4	Warp Yarn, Weft Yarn, Size
T5	Reactive Dye, Salt
T6	Warp Yarn, Size
T7	Reactive Dye, Soda Ash
T8	Salt, Soda Ash

Item counts: Warp Yarn 4, Size 4, Reactive Dye 3, Salt 3, Soda Ash 3, Weft Yarn 2 (supports 50 %, 50 %, 37.5 %, 37.5 %, 37.5 %, 25 %). The candidate rules:

RULE	SUPPORT	CONFIDENCE	VERDICT (MIN SUP 25 %, MIN CONF 60 %)
Warp Yarn → Size	{both} = 4/8 = **50.0 %**	4 ÷ 4 = **100 %**	**Survives — the strongest rule**
Size → Warp Yarn	50.0 %	4 ÷ 4 = **100 %**	Survives
Reactive Dye → Salt	{both} = 2/8 = **25.0 %**	2 ÷ 3 = **66.7 %**	Survives
Salt → Reactive Dye	25.0 %	2 ÷ 3 = **66.7 %**	Survives (note: not equal to the reverse rule in general — here the counts coincide)
Reactive Dye → Soda Ash	2/8 = 25.0 %	2 ÷ 3 = 66.7 %	Survives — the dyehouse pair

Reading the result: every warp-yarn issue took size with it (confidence 100 % at 50 % support) — sizing preparation and warp issue are one operation in this mill, so stores should kit them together and the size stock level should be planned directly from the warp-yarn programme. The dye rules are weaker (66.7 %) — dye is sometimes issued with salt, sometimes with soda ash alone — which

matches exhaust dyeing practice, where salt and alkali enter at different stages and may be drawn on different requisitions. The mining has recovered the mill's real process structure from issue slips alone.

5. Practice tasks

1. A ninth transaction is added: T9 = {Reactive Dye, Salt}. Recompute (a) the support of {Reactive Dye + Salt}, and (b) the confidence of Reactive Dye → Salt.
2. State what happens to the rule Reactive Dye → Soda Ash if minimum support is raised to 40 %, and what that demonstrates about mining thresholds.
3. A manager reads Warp Yarn → Size (100 %) and orders the store to issue size automatically with every warp-yarn requisition, doubling size consumption in a month when sizing ran normally. Identify the flaw in the action taken.

Answer key

1. With N = 9: {Dye + Salt} now occurs in T3, T5, T9 = 3 transactions → support = 3 ÷ 9 = **33.3 %**. Dye now occurs 4 times, and 3 of those include salt → confidence = 3 ÷ 4 = **75 %**. One extra transaction has strengthened both measures — small datasets move fast, which is why mining conclusions are re-checked as data grows.
2. The rule's support is 25 %, below the 40 % threshold, so it is **pruned** — it disappears from the output although nothing about the mill changed. Thresholds filter evidence, they do not create or destroy facts: a high min-support hides every rare pattern, including rare faults, so thresholds must be chosen for the purpose, not by habit.
3. The rule describes **co-issue in the records**, not a required consumption ratio. Warp yarn and size travel together because sizing follows warping in the process — the size actually consumed is set by the sizing recipe and the warp length, not by the number of requisitions. Auto-issuing per requisition double-counts (several requisitions may serve one sizing programme) and confuses correlation with a dosing rule. The correct use of the rule is planning — stock and kitting — while quantities remain governed by the process recipe.
